

HOUSE PRICE PREDICTION PROJECT



Introduction

House Price Prediction is a Machine Learning project used to estimate the price of houses based on various features. It is a regression problem where the output is a continuous value. This project helps in understanding how different factors like size, location, and condition affect house prices. Machine Learning models analyze patterns in historical data to make predictions. It is widely used in real estate industries. This project demonstrates how data-driven approaches can improve decision-making. Various algorithms are used to compare performance. The main goal is to build an accurate prediction model. It also helps in learning data preprocessing and model evaluation techniques.

Dataset Description

The dataset used is **kc_house_data.csv**, which contains housing-related information. It includes multiple features describing the properties of houses. Each row represents a single house. The dataset contains both numerical and categorical data. Important attributes include number of bedrooms, bathrooms, square footage, and year built. The dataset is widely used for regression analysis. It provides a good understanding of real-world data. Some features may require cleaning or transformation. The dataset size is large enough for training models. It is suitable for comparing multiple algorithms.

Input and Output Variables

Input Features (X):

- bedrooms
- bathrooms
- sqft_living
- sqft_lot
- floors
- condition
- grade
- sqft_basement
- yr_built
- yr_renovated

Output Variable (y):

- price

These input features are used to predict the house price. Each feature contributes differently to the final output.

Data Preprocessing

Data preprocessing is an important step in Machine Learning. It involves cleaning and preparing data for model training. The dataset is loaded using Pandas. Missing values are checked and handled if necessary. Features and target variables are separated. Data is then split into training and testing sets. This ensures the model is evaluated on unseen data. Preprocessing improves model accuracy. It also removes inconsistencies in the dataset. Proper preprocessing is essential for reliable results. It forms the foundation of any ML project.

Feature Scaling

Feature scaling is used to standardize the range of independent variables. Different features may have different scales. For example, square footage values are much larger than the number of bedrooms. This can bias the model. StandardScaler is used to normalize the data. It transforms data so that mean is 0 and standard deviation is 1. Scaling improves model performance. It is especially important for algorithms like SVR and KNN. It ensures all features

contribute equally. It speeds up convergence during training. It leads to more accurate predictions.

Machine Learning Algorithms Used

1. Linear Regression

Linear Regression is a basic algorithm used for predicting continuous values. It assumes a linear relationship between input variables and output. The model fits a straight line that minimizes error. It is easy to understand and implement. It works well when data follows a linear pattern. It helps identify how features influence the price. However, it cannot handle complex relationships. It is sensitive to outliers. It is often used as a baseline model. Despite limitations, it provides useful insights.

2. Ridge Regression

Ridge Regression improves Linear Regression by adding regularization. It reduces overfitting by penalizing large coefficients. This helps in handling multicollinearity. It keeps all features but reduces their impact. The model becomes more stable. It performs well with complex datasets. It balances bias and variance. It improves generalization on test data. It is useful when many features are present. It enhances prediction accuracy.

3. Lasso Regression

Lasso Regression also uses regularization but with absolute values. It can shrink some coefficients to zero. This means it performs feature selection. It removes less important features automatically. It simplifies the model. It improves interpretability. It is useful when dataset has many irrelevant features. It reduces overfitting. It helps in building efficient models. However, it requires proper tuning.

4. Decision Tree Regressor

Decision Tree splits data into smaller parts based on feature values. It creates a tree-like structure. Each node represents a decision. It can model complex relationships. It does not require scaling. It is easy to visualize. However, it may overfit the data. Small changes can affect structure. Pruning helps improve performance. It is flexible and powerful.

5.Random Forest Regressor

Random Forest uses multiple decision trees. It combines their outputs for better accuracy. It reduces overfitting. It handles large datasets well. It captures complex patterns. It is more reliable than a single tree. It provides feature importance. It works well in most cases. It requires more computation. It is widely used in real-world problems.

6.Support Vector Regressor (SVR)

SVR is used for regression tasks. It fits a model within a margin of error. It handles high-dimensional data well. It uses kernel functions for non-linear data. It requires feature scaling. It is sensitive to parameter tuning. It performs well in complex scenarios. However, it can be slow. It provides accurate predictions when tuned properly.

7. K-Nearest Neighbors (KNN) Regressor

KNN predicts values based on nearest data points. It uses distance to find neighbors. It is simple and easy to understand. It does not require training. It works well with small datasets. It captures local patterns. It is sensitive to K value. Scaling is important for accuracy. It becomes slow for large datasets. It is useful for basic prediction tasks.

Model Evaluation

Models are evaluated using regression metrics. R^2 Score measures how well the model fits the data. RMSE measures prediction error. Lower RMSE means better performance. Models are trained on training data. They are tested on unseen data. This ensures reliability. Evaluation helps compare algorithms. It identifies the best model. Proper evaluation is essential.

Results

Different algorithms produce different results. Random Forest usually gives the best accuracy. Linear models perform well for simple data. SVR and KNN work better with proper tuning. The Decision Tree may overfit. Regularized models improve stability. Results depend on data quality. Overall, ensemble methods perform better.

Conclusion

This project demonstrates how Machine Learning can predict house prices. Multiple algorithms were applied and compared. Random Forest performed best in most cases. Feature scaling improved accuracy. Proper preprocessing was important. Each algorithm has strengths and limitations. Choosing the right model is crucial. This project provides practical ML experience. It shows real-world application of regression.

What I Learned

- Understanding regression problems
- Data preprocessing techniques
- Feature scaling importance
- Working of different ML algorithms
- Model training and prediction
- Evaluation using R^2 and RMSE
- Comparing model performance
- Handling real-world datasets
- Improving accuracy using techniques
- Writing structured ML code